

Problem Set #2 : Sedra Khan

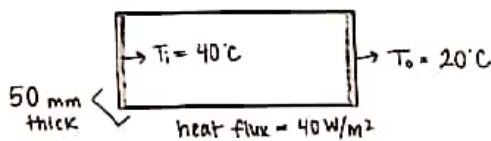
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Jan. 27th, 2020.
CHE 312

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Do problems 1.6, 1.7, 1.14, and 1.27 from our textbook (Bergman et al.; 2019), on pages P-1, P-2 and P-4.

- 1.6 The heat flux through a wood slab 50 mm thick, whose inner and outer surface temperatures are 40 and 20°C, respectively, has been determined to be 40 W/m². What is the thermal conductivity of the wood?



What's the thermal conductivity of the wood (K)?

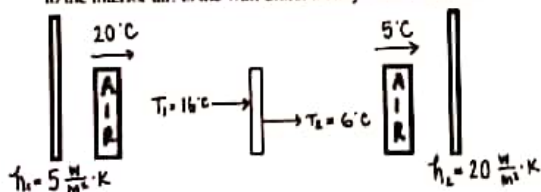
$$\text{heat flux } q'' \left(\frac{\text{W}}{\text{m}^2} \right) = -K \frac{T_2 - T_1}{L}$$

$$= -K \frac{\Delta T}{L}$$

$$40 \text{ W/m}^2 = -K \frac{(20 - 40)^\circ\text{C}}{0.05 \text{ m}}$$

$$\therefore K = .10 \left(\frac{\text{W}}{\text{m}\cdot\text{K}} \right)$$

- 1.14 A wall has inner and outer surface temperatures of 16 and 6°C, respectively. The interior and exterior air temperatures are 20 and 5°C, respectively. The inner and outer convection heat transfer coefficients are 5 and 20 W/m²·K, respectively. Calculate the heat flux from the interior air to the wall, from the wall to the exterior air, and from the wall to the interior air. Is the wall under steady-state conditions?



interior air to the wall:

$$q'' = h(T_s - T_\infty) = 5(20 - 16) = 20 \text{ W/m}^2$$

wall to the exterior:

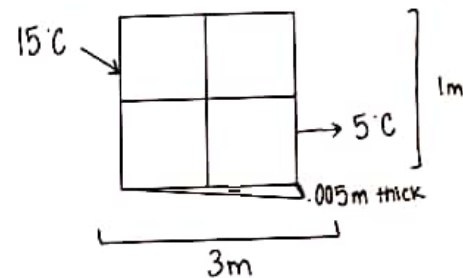
$$q'' = h(T_s - T_\infty) = 20(6 - 5) = 20 \text{ W/m}^2$$

wall to the interior:

$$q'' = h(T_s - T_\infty) = 5(16 - 20) = -20 \text{ W/m}^2$$

Yes, it is under steady state equation

- 1.7 The inner and outer surface temperatures of a glass window 5 mm thick are 15 and 5°C. The thermal resistance of the glass window due to conduction is $R_{t, \text{cond}} = 1.19 \times 10^{-3} \text{ K/W}$. What is the rate of heat loss through a 1 m × 3 m window? What is the thermal conductivity of the glass?



$$R_{t, \text{cond}} = 1.19 \times 10^{-3} \text{ K/W}$$

heat loss? K?

↳ according to the thermal resistance concept:

$$R_{t, \text{cond}} = \frac{L}{KA}$$

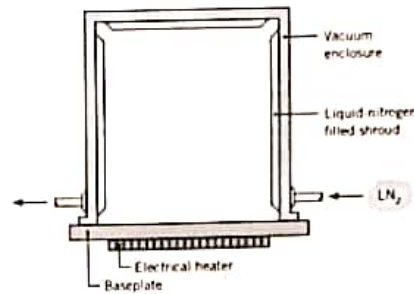
$$\therefore K = \frac{L}{A \cdot R_{t, \text{cond}}} = \frac{0.005 \text{ m}}{3 \text{ m}^2 (1.19 \times 10^{-3} \text{ K/W})}$$

$$\therefore K = 1.4 \text{ W/K}\cdot\text{m}$$

$$\text{heat loss} = q = q'' A = \frac{\Delta T}{R_T} \quad (1.11)$$

$$\therefore q = \frac{(15 - 5) \text{ K}}{1.19 \times 10^{-3} \text{ K/W}} = 8403 \text{ W}$$

1.27 A vacuum system, as used in sputtering electrically conducting thin films on microcircuits, is comprised of a baseplate maintained by an electrical heater at 400 K and a shroud within the enclosure maintained at 97 K by a liquid-nitrogen coolant loop. The circular baseplate, insulated on the lower side, is 0.3 m in diameter and has an emissivity of 0.25.



- How much electrical power must be provided to the baseplate heater?
- At what rate must liquid nitrogen be supplied to the shroud if its heat of vaporization is 125 kJ/kg?
- To reduce the liquid nitrogen consumption, it is proposed to bond a thin sheet of aluminum foil ($\epsilon = 0.09$) to the baseplate. Will this have the desired effect?

(a) electrical power? $q''_{\text{rad}} = \epsilon \sigma (T_s^4 - T_{\text{sur}}^4)$ — $E = \epsilon \sigma T_s^4$
 Stephan boltzman constant = $5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \cdot \text{K}^4}$

$$q'' = (0.25)(5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \cdot \text{K}^4})(400^4) \text{ K}^4 = 361.6 \text{ W/m}^2$$

$$q = q'' A = (361.6 \frac{\text{W}}{\text{m}^2}) (\pi (\frac{0.30}{2})^2) = \boxed{25.6 \text{ W}}$$

(b) rate of liquid nitrogen?

$$q = \dot{m} \cdot \underbrace{c_p (T_{\text{out}} - T_{\text{in}})}_{\text{heat of vaporization}}$$

$$\dot{m} = \frac{q}{\text{heat of Vap.}} = \frac{25.6 \text{ W}}{125 \text{ kJ/kg}} = \frac{0.0256 \text{ KJ/s}}{125 \text{ kJ/kg}} = \boxed{2.04 \times 10^{-4} \frac{\text{kg}}{\text{s}}}$$

(c) $\epsilon = 0.09$?

if the emissivity is lowered q will decrease

$$\hookrightarrow q = \epsilon \cdot \sigma \cdot (T_s^4 - T^4) \cdot A$$

if q decreases the flow rate, $\dot{m}$, will also decrease

$$\hookrightarrow \dot{m} = q / \text{heat of vaporization}$$

$\therefore$ Yes, it **WILL** have the desired effect